

SHAD ALI SHAH

Economist | Data Scientist | Data Analyst | Business Intelligence Analyst

Islamabad, Pakistan | GCC Relocation | Full-Time Available | +92-334-890-1607 | shadalishah45@gmail.com |

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)



Professional Summary

Entry-level Economics and Data professional with **MPhil Economics coursework and thesis completed**, a **Data Science & Analytics specialisation**, Graduate Research Assistant experience, three analytics internships, and 30+ multidisciplinary projects. Proficient in **Python, SQL, R, Stata, EViews, Power BI, and Tableau**, with experience in econometrics, data analysis, statistical modelling, machine learning, forecasting, and analytical reporting.

Core Competencies

Economics Research	&	Macroeconomics, microeconomics, development economics, financial economics, policy analysis, literature review, quantitative research
Econometrics Statistics	&	VAR, SVAR, VECM, ARIMA, ADF/KPSS tests, Johansen cointegration, Panel Data (PMG-ARDL), Granger causality, IRFs, FEVD, statistical modelling
Data Science & ML		Deep Learning (LSTM, BERT), Python, Pandas, NumPy, Scikit-learn, regression, classification, Random Forest, K-Means, feature engineering, model evaluation
Analytics & BI		SQL, Google BigQuery, Excel, Power BI, Tableau, data cleaning, EDA, data visualisation, forecasting, KPI reporting, dashboard development
Tools		R, Stata, EViews, Jupyter Notebook, Google Colab, Git/GitHub, Quarto, LaTeX

Professional Experience

Data Trainer

SME Careers

July 2026 - Present

Remote

- Create and review prompts and AI responses to improve training quality, relevance, and consistency across different topics.
- Evaluate and rank AI-generated responses to identify quality issues and improve model performance across varied contexts.
- Test AI outputs for inaccuracies and potential bias, supporting reliable and context-appropriate model responses.

Graduate Research Assistant — Economics & Data Science

Quaid-i-Azam University

February 2026 - Present

Islamabad

- Review economic literature and synthesise evidence for faculty-supervised research, producing structured references and findings summaries.
- Collect, clean, validate, and maintain research datasets using Python and Google Colab, supporting accurate quantitative and econometric analysis.
- Analyse research data and produce tables, visualisations, and structured findings while maintaining reproducible analytical workflows.

Data Science Intern

Arch Technologies

1 May 2026 - 30 June 2026

Remote

- Built and compared Logistic Regression and Random Forest classifiers on 891 passenger records; Random Forest achieved **81.56% accuracy** and 78.22% F1-score.
- Forecasted Apple trading prices using 2,517 observations with Linear Regression and LSTM; Linear Regression achieved **99.2% R-squared**, \$2.64 MAE, and \$3.71 RMSE.
- Segmented 1,000 customer records into **six interpretable groups** using K-Means, supporting customer profiling and behavioural analysis.

Data Science Intern

CodeAlpha

10 May 2026 - 10 June 2026

Remote

- Developed advertising-sales prediction models using 200 observations; Random Forest achieved **98.13% R-squared**, 0.6201 MAE, and 0.7686 RMSE.
- Compared Linear Regression, Random Forest, and Gradient Boosting on 301 used-car records; Linear Regression achieved the strongest result at **75.27% R-squared**.
- Cleaned, validated, encoded, and standardised data using Python and Pandas, producing reproducible datasets and documented model comparisons.

AI/ML Engineering Intern - Best Intern Award

DevelopersHub Corporation

11 May 2026 - 22 June 2026

Remote

- Analysed **120,000 news articles** using text preprocessing, n-grams, and Named Entity Recognition to identify relevant textual patterns.
- Designed a BERT-based classification workflow and evaluation plan using a 7,600-article test set.
- Documented NLP, classification, regression, and RAG workflows; earned the Best Intern Award for analytical performance, documentation, and timely delivery.

Independent Economics, Data Science & Analytics Projects | [GitHub](#) | 2025-2026

- Conducted self-directed analysis across macroeconomic, employment, financial, customer, and transaction datasets, producing **30+ documented projects**.
- Applied econometric, statistical, and machine-learning methods to generate forecasts, identify relationships, classify observations, and segment customers.
- Used Python, R, SQL, Stata, EViews, BigQuery, Power BI, and Tableau to produce analytical models, KPI analyses, dashboards, and visualisations.
- Developed reproducible analytical workflows, notebooks, dashboards, research summaries, and documented GitHub repositories.

Selected Economics, Data Science & Analytics Projects

Pakistan Macroeconomic Dynamics

GitHub

Python, statsmodels, VAR, SVAR, VECM, ARIMA, Quarto

- Analysed Pakistan's 2010–2020 quarterly GDP, spending, revenue, inflation, and interest-rate data to assess fiscal-monetary linkages.
- Applied ADF/KPSS, ARIMA/ARMA, VAR, Granger causality, IRFs, FEVD, and short-run identified SVAR in Python.
- Identified mixed I(0)/I(1) series and no cointegration, supporting first-differenced VAR; delivered a reproducible policy-analysis report.

Pakistan E-Commerce Analytics

GitHub

Python, RFM, K-Means, Prophet, LightGBM, XGBoost, SHAP

- Cleaned 1,048,575 raw rows into **584,524 records** with 32 engineered features and analysed 115,326 customers.
- Applied RFM/K-Means segmentation, Prophet/LightGBM forecasting, and XGBoost-SHAP churn modelling to identify customer and sales patterns.
- Segmented 60,367 customers into six RFM groups; Prophet achieved **35.9% MAPE** versus 46.3% for LightGBM.

ComplaintIQ AI — Consumer Complaint Intelligence System

GitHub

Python, Pandas, NumPy, NLTK, Scikit-learn, NLP

- Analysed **162,000+ consumer complaints** across five product categories to identify customer intents and pain points.
- Applied NLP preprocessing, n-gram analysis, intent tagging, and class-imbalance diagnostics for automated classification.
- Compared ML, deep learning, and LLM approaches and designed a triage architecture with urgency flagging for high-risk complaints.

Supply Chain Performance Dashboard

GitHub

Power BI, Power Query, DAX, Business Intelligence

- Analysed 100 SKUs, 46K units sold, and **577.6K in reported revenue** across inventory, quality, and supplier performance.
- Used Power Query and DAX to transform data and develop KPIs for revenue, inventory, defects, and inspections.
- Identified 17 low-stock SKUs, 2.28% average defects, and **61.0% inspection failures**, informing inventory and supplier decisions.

Education

MPhil Economics - Coursework and Thesis Completed | Degree Completion Pending

2025 - Jan 2027

Quaid-i-Azam University, Islamabad

CGPA: 3.50/4.00

Completed all MPhil coursework and thesis under faculty supervision. Thesis approved by supervisor; formal degree completion is pending. Focus: Econometrics, Quantitative Research, Time-Series Analysis, Data Science, and Macroeconomic Analysis. **Available for full-time employment.**

BS Economics - Graduated with Distinction

2020 - 2024

Kohat University of Science and Technology

CGPA: 3.89/4.00

Relevant Coursework: Econometrics, Statistics, Mathematical Economics, Macroeconomics, Microeconomics, Development Economics, Monetary Economics, Financial Institutions and Markets, Public Finance and Fiscal Policy, Research Methodology

Undergraduate Thesis: The Influence of Biofuel Consumption on Environmental Sustainability in Developed Countries
Method & Result: PMG-ARDL; biofuel consumption had a beneficial long-run effect on environmental sustainability.

Certifications

- IBM Data Science Professional Certificate; IBM Data Analyst Professional Certificate – IBM
- Microsoft Power BI Data Analyst; Microsoft AI & ML Engineering – Microsoft
- Google Advanced Data Analytics; Google Business Intelligence; Google Data Analytics – Google
- Machine Learning Specialisation – DeepLearning.AI & Stanford Online
- Generative AI Data Analyst – Vanderbilt University
- Survey Data Collection and Analytics – University of Michigan
- Python and Statistics for Financial Analysis – HKUST